

Software Engineering	
Course Code: IS52	Credits: 2:0:1:2
Pre – requisites:	Contact Hours: 30L+15T+30S
Course Coordinator: Dr. Krishna Raj RM	

Course Content

Unit I

Introduction: Professional software development; Software engineering ethics; Case studies. Software processes: Software process models; Process activities; Coping with change; Process improvement. Agile software development: Agile methods; Plan-driven and agile development; Extreme programming; Agile project management; Scaling agile methods.

Lab Exercises:

Initialize a GitHub repository and set up a Kanban board to manage tasks for a sample project. Define a Product Backlog and perform a mini-sprint including task estimation and daily updates.

- **Pedagogy:** Lecture with case-based discussion
- **Links:** <https://nptel.ac.in/courses/106105182>

Unit II

Requirements engineering: Functional and non-functional requirements; The software requirements document; Requirements specification; Requirements engineering processes; Requirements elicitation and analysis; Requirements validation; Requirements management. System modeling: Context models; Interaction models; Structural models; Behavioral models; Model-driven engineering.

Lab Exercises:

Develop a Software Requirements Specification (SRS) document for a given problem statement. Draw Use Case and Activity diagrams for a chosen system using a tool like StarUML or Draw.io.

- **Pedagogy:** Problem-based learning and modeling exercises
- **Links:** <https://nptel.ac.in/courses/106105182>

Unit III

Architectural design: Architectural design decisions; Architectural views; Architectural patterns; Application architectures. Design and implementation: Object-oriented design using the UML; Design patterns; Implementation issues; Open source development.

Lab Exercises:

Create detailed Class and Sequence diagrams to represent the internal logic of the system. Map system requirements to a specific architectural pattern.

- **Pedagogy:** Design-oriented discussions and worked examples
- **Links:** <https://nptel.ac.in/courses/106105182>

Unit IV

Software testing: Development testing; Test-driven development; Release testing; User testing. Software evolution: Evolution processes; Legacy systems; Software maintenance.

Lab Exercises:

Write and execute test cases for core modules using frameworks like JUnit or PyTest.

Set up a simple automated build pipeline (e.g., GitHub Actions) to run tests on every code push.

- **Pedagogy:** Demonstration-based learning with illustrative examples
- **Links:** <https://nptel.ac.in/courses/106105182>

Unit V

Project management: Management activities; Project planning; Risk management; Project scheduling. Project planning: Software pricing; Plan-driven development; Project scheduling; Estimation techniques. Quality management: Software quality; Software standards; Reviews and inspections; Software measurement and metrics.

Lab Exercises:

Develop a complete project plan for a given case study.

Run a static code analysis tool to identify "code smells" and technical debt in your project.

- **Pedagogy:** Case-study based learning and numerical problem solving
- **Links:** <https://nptel.ac.in/courses/106105182>

Text Book:

1. Ian Sommerville, Software Engineering, 10th Edition, Pearson, 2020.

Reference Books:

1. Roger S. Pressman and Bruce R. Maxim, Software Engineering: A Practitioner's Approach, 9th Edition, McGraw-Hill, 2020.
2. Len Bass, Paul Clements, Rick Kazman, Software Architecture in Practice, 4th Edition, Addison-Wesley, 2021

Self-Learning: Students shall carry out a hands-on exercise such as preparing a Software Requirements Specification (SRS) and UML models for a given problem statement. It may also include reviewing an open-source project to understand its software process, design structure and version control practices.

Course Outcomes(COs):

At the end of the course, the student will be able to:

1. Apply the concepts of software engineering and agile development processes using modern project management tools. (PO1, 5, 8, 9, 10, 11) (PSO 1)
2. Analyze functional and non-functional requirements to create comprehensive software specifications and system models. (PO1, 2, 5, 9, 11) (PSO 1)
3. Design architectural patterns and object-oriented systems using modeling tools and design principles. (PO1, 2, 3, 5, 9) (PSO 1)
4. Implement software testing strategies and automated build pipelines to manage software evolution. (PO1, 4, 5, 11) (PSO 1)
5. Evaluate software project plans, quality metrics and ethics through risk management and static analysis. (PO1, 2, 5, 7, 10) (PSO 1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		

Other Components		
Tool Demonstration (Jira Tool)	10	CO1, CO2, CO3, CO4,CO5
Case Study	10	CO1, CO2, CO3, CO4,CO5
Semester-End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5